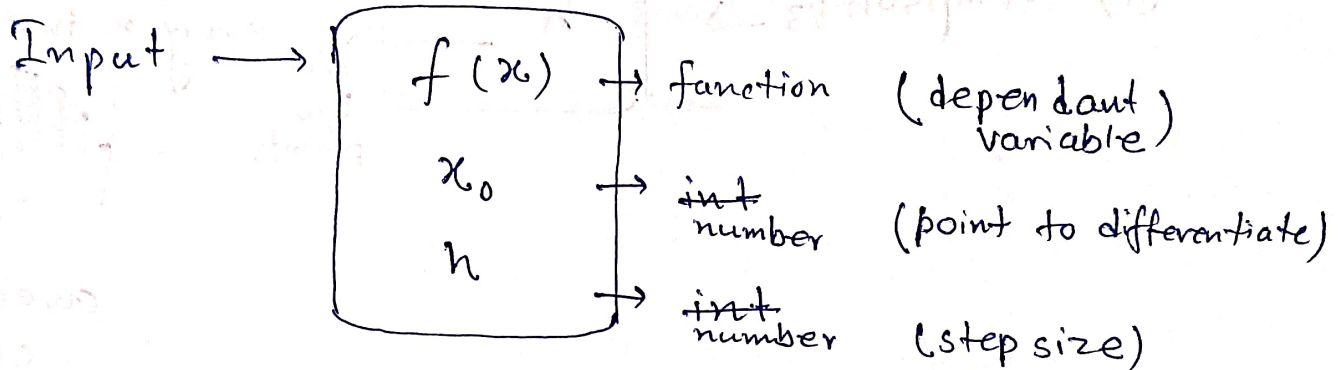


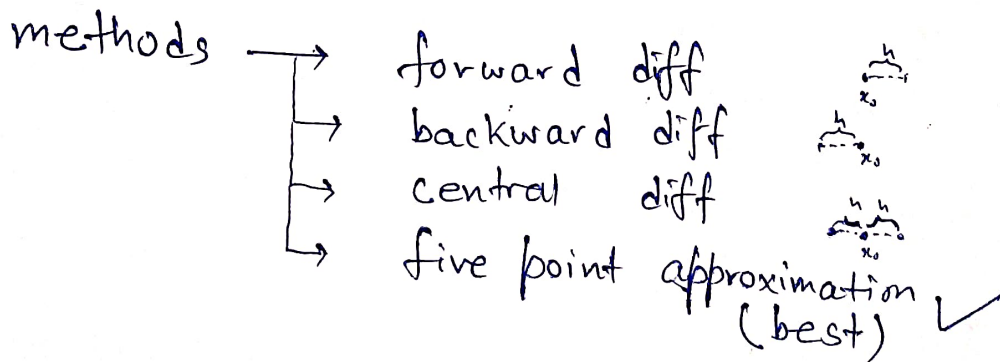
Computational Physics

(3) →

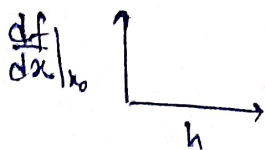
Differentiation: gives $\frac{df(x)}{dx}$ at a point x_0



Output → number $\left(\frac{df}{dx} \Big|_{x=x_0} \right)$
(slope at x_0)



error plot :



x_arr → array of different h

y_arr → array of $\left| \text{slope at } x_0 - \text{exact slope at } x_0 \right|$ for all h

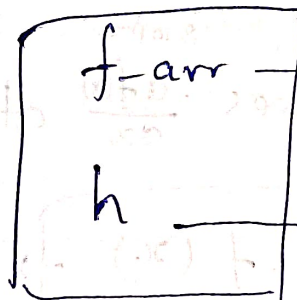
Integration :

methods:

(i) trapezoidal $\rightarrow$

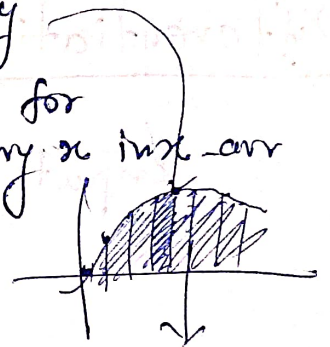
(ii) simpson $1/3 \rightarrow$

Input



array of $f(x)$ for every x in arr

dist b/w points



area under curve
 $\downarrow$
output

(4) ODE

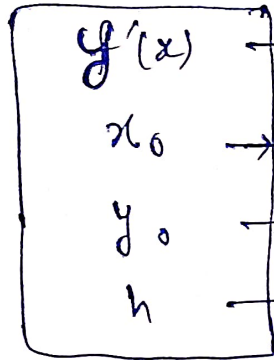
Initial Value Problems :

given: $y'(y, x) ; y(x)$
 ~~$y(x), f(x), y'(x), y(x), y(x)$~~

Methods

- Euler:
- midpoint(RK2)
- RK4

Input



type

function

number

number

step number

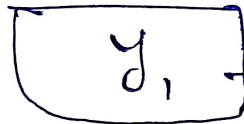
desc.

(the ODE / first order integral in terms of x & $y(x)$)

points to start approximation

step size for each approximation next

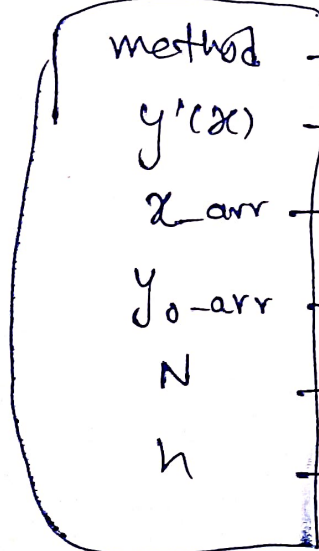
output



number

→ $y(x+h)$
y value next point of

special $f'' \rightarrow$ Caller Input



function

→ the solver method

function

→ ODE / first order

num array

→ $[x_0, x_1, x_2, \dots, x_N]$

num array

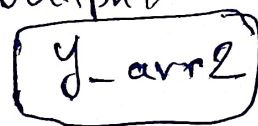
→ $[y_{0A}, y_{0B}, y_{0C}, \dots]$ init y values

number

number

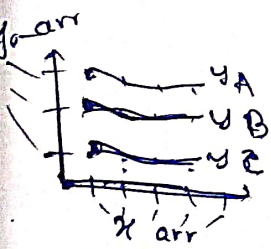
→ step size

output



→ 2D num array

→ $\begin{bmatrix} [y_{0A}, y_{0B}, \dots, (y_{0-arr})] \\ [y(x_1), y(x_2), \dots, y(x_N)] \\ \vdots \\ [y(x_N)] \end{bmatrix}$



(must be spaced h) and due to x_N

$y_s = \begin{bmatrix} y_{0A} \\ y_{0B} \\ \vdots \\ y_{0N} \end{bmatrix}$